

## TRI-COUNTY REGIONAL, WI, USA

WMO: 726416

Lat: 43.212N

Lon: 90.181W

Elev: 717

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 14921

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.9      | -5.0 | -17.6                         | 2.2 | -8.7 | -12.9 | 2.8 | -3.8 | 23.8                  | 21.4 | 20.4 | 20.7 | 3.5                   | 260  | 0.524 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.1                   | 90.2            | 74.4 | 87.6 | 73.1 | 84.6 | 71.2 | 77.4                | 86.5 | 75.5 | 84.0 | 73.8 | 81.7 | 9.6                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.5                            | 132.4 | 83.0 | 72.7 | 124.3 | 80.9 | 71.1 | 118.0 | 79.2 | 41.2          | 86.4 | 39.5 | 84.4 | 37.7 | 81.7 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |      |
|-------------------|------|------|----------------------------|-------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
| Min               |      | Max  |                            | Min   | Max   |                    | Min                                                | Max       | Min   | Max        | Min   | Max        | Min   | Max        |       |      |
| (a)               | (b)  | (c)  |                            | (e)   | (f)   | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 21.1              | 18.7 | 16.7 |                            | -20.4 | 94.4  | 6.6                | 3.1                                                | -25.2     | 96.6  | -29.0      | 98.4  | -32.7      | 100.2 | -37.5      | 102.4 |      |
|                   |      |      |                            | DB    |       |                    |                                                    |           |       |            |       |            |       |            |       |      |
|                   |      |      |                            | WB    | -20.3 | 80.5               | 5.6                                                | 1.8       | -24.3 | 81.8       | -27.5 | 82.9       | -30.7 | 83.9       | -34.7 | 85.2 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 47.5   | 19.2  | 21.7  | 35.0  | 47.7  | 59.0  | 68.6  | 72.3  | 70.1  | 63.0  | 49.9  | 37.4  | 24.4  |
|                                                                                  | DBStd       | 20.99  | 12.49 | 12.15 | 11.79 | 9.51  | 8.65  | 6.32  | 5.53  | 5.47  | 7.90  | 9.14  | 9.74  | 12.00 |
|                                                                                  | HDD50       | 3709   | 955   | 792   | 485   | 151   | 18    | 0     | 0     | 0     | 5     | 117   | 392   | 794   |
|                                                                                  | HDD65       | 7064   | 1420  | 1211  | 932   | 522   | 228   | 34    | 7     | 16    | 129   | 478   | 828   | 1259  |
|                                                                                  | CDD50       | 2795   | 0     | 0     | 20    | 83    | 297   | 560   | 690   | 623   | 394   | 113   | 14    | 1     |
|                                                                                  | CDD65       | 675    | 0     | 0     | 2     | 4     | 43    | 144   | 232   | 173   | 69    | 8     | 0     | 0     |
|                                                                                  | CDH74       | 6529   | 0     | 0     | 15    | 76    | 485   | 1393  | 2278  | 1519  | 675   | 88    | 0     | 0     |
|                                                                                  | CDH80       | 2053   | 0     | 0     | 1     | 14    | 141   | 439   | 803   | 449   | 192   | 14    | 0     | 0     |
| Wind                                                                             | WSAvg       | 6.8    | 7.3   | 7.7   | 7.9   | 9.0   | 7.6   | 6.1   | 4.9   | 4.7   | 5.5   | 6.9   | 7.2   | 7.2   |
| Precipitation                                                                    | PrecAvg     | 35.4   | 0.9   | 1.2   | 2.0   | 3.9   | 4.1   | 5.7   | 4.2   | 4.5   | 3.3   | 2.2   | 2.1   | 1.7   |
|                                                                                  | PrecMax     | 48.3   | 2.0   | 2.5   | 4.3   | 6.3   | 10.8  | 10.9  | 8.9   | 18.2  | 7.8   | 4.9   | 6.3   | 3.7   |
|                                                                                  | PrecMin     | 24.0   | 0.3   | 0.4   | 0.6   | 1.5   | 1.5   | 1.3   | 1.7   | 1.1   | 0.6   | 0.7   | 0.3   | 0.2   |
|                                                                                  | PrecStd     | 6.3    | 0.5   | 0.6   | 1.2   | 1.5   | 2.0   | 2.5   | 2.0   | 3.9   | 1.8   | 1.2   | 1.5   | 0.9   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.8  | 56.8  | 75.8  | 81.8  | 90.1  | 92.3  | 95.9  | 91.4  | 89.5  | 81.9  | 69.6  | 57.2  |
|                                                                                  |             | MCWB   | 42.3  | 49.6  | 62.5  | 62.7  | 71.3  | 74.7  | 76.4  | 75.3  | 73.1  | 66.6  | 57.8  | 54.3  |
|                                                                                  | 2%          | DB     | 43.1  | 46.5  | 64.6  | 75.0  | 83.8  | 88.3  | 90.3  | 88.2  | 85.4  | 75.6  | 62.8  | 48.1  |
|                                                                                  |             | MCWB   | 38.0  | 41.1  | 54.7  | 59.7  | 66.7  | 72.7  | 75.5  | 74.1  | 71.5  | 64.2  | 53.8  | 44.2  |
|                                                                                  | 5%          | DB     | 38.5  | 41.6  | 58.8  | 70.3  | 79.5  | 84.8  | 87.5  | 84.8  | 81.5  | 71.0  | 57.5  | 43.4  |
|                                                                                  |             | MCWB   | 35.6  | 37.1  | 50.5  | 56.4  | 64.8  | 70.6  | 73.8  | 71.9  | 68.4  | 61.5  | 50.0  | 40.0  |
|                                                                                  | 10%         | DB     | 35.8  | 37.2  | 52.0  | 65.4  | 75.0  | 81.8  | 84.3  | 81.9  | 77.7  | 65.9  | 53.0  | 39.2  |
|                                                                                  |             | MCWB   | 33.4  | 33.5  | 44.6  | 53.1  | 62.5  | 69.2  | 72.1  | 70.4  | 66.2  | 57.4  | 46.9  | 36.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 43.0  | 48.6  | 63.4  | 65.3  | 73.3  | 78.1  | 80.2  | 78.7  | 76.2  | 71.3  | 59.3  | 55.0  |
|                                                                                  |             | MCDB   | 46.8  | 55.5  | 75.2  | 76.1  | 86.5  | 87.6  | 89.7  | 87.8  | 85.7  | 77.6  | 66.5  | 57.8  |
|                                                                                  | 2%          | WB     | 38.6  | 41.9  | 57.8  | 61.4  | 70.0  | 75.5  | 77.9  | 76.4  | 73.3  | 67.3  | 55.5  | 44.4  |
|                                                                                  |             | MCDB   | 42.1  | 45.8  | 63.4  | 72.5  | 79.6  | 84.5  | 87.3  | 84.4  | 81.4  | 72.4  | 61.5  | 46.7  |
|                                                                                  | 5%          | WB     | 35.8  | 37.4  | 50.5  | 58.5  | 67.4  | 73.2  | 76.0  | 74.4  | 70.9  | 63.1  | 51.3  | 40.2  |
|                                                                                  |             | MCDB   | 38.6  | 41.2  | 57.6  | 68.3  | 76.0  | 82.5  | 84.9  | 81.7  | 78.4  | 69.3  | 56.6  | 43.2  |
|                                                                                  | 10%         | WB     | 33.3  | 34.0  | 45.3  | 54.4  | 64.5  | 70.9  | 74.2  | 72.5  | 68.7  | 58.4  | 46.9  | 36.5  |
|                                                                                  |             | MCDB   | 35.5  | 36.7  | 51.6  | 64.1  | 72.4  | 79.0  | 81.9  | 79.5  | 75.1  | 64.8  | 51.7  | 39.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 17.1  | 18.0  | 18.7  | 21.6  | 21.5  | 20.6  | 21.1  | 20.8  | 22.5  | 20.3  | 17.4  | 15.5  |
|                                                                                  |             | MCDBR  | 18.7  | 21.0  | 25.6  | 29.6  | 27.0  | 24.8  | 24.0  | 23.5  | 26.4  | 26.9  | 25.2  | 18.1  |
|                                                                                  |             | MCWBR  | 14.9  | 16.0  | 16.2  | 16.8  | 12.9  | 11.9  | 11.1  | 11.4  | 13.5  | 16.0  | 17.6  | 15.0  |
|                                                                                  | 5% WB       | MCDBR  | 17.2  | 20.5  | 23.5  | 25.2  | 22.1  | 20.9  | 20.2  | 19.5  | 21.3  | 22.5  | 22.9  | 17.0  |
|                                                                                  |             | MCWBR  | 14.2  | 16.3  | 16.1  | 15.9  | 12.2  | 11.4  | 11.1  | 10.7  | 12.1  | 15.9  | 17.6  | 15.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.274  | 0.308 | 0.326 | 0.372 | 0.399 | 0.419 | 0.422 | 0.414 | 0.376 | 0.336 | 0.296 | 0.275 |       |
|                                                                                  | taud        | 2.383  | 2.290 | 2.391 | 2.306 | 2.280 | 2.261 | 2.257 | 2.278 | 2.367 | 2.476 | 2.512 | 2.438 |       |
|                                                                                  | Ebn at Noon | 275    | 280   | 291   | 283   | 277   | 271   | 269   | 267   | 270   | 269   | 266   | 265   |       |
|                                                                                  | Edh at Noon | 24     | 32    | 33    | 39    | 42    | 43    | 42    | 40    | 34    | 27    | 22    | 21    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 557    | 890   | 1216  | 1440  | 1709  | 1901  | 1930  | 1669  | 1364  | 879   | 577   | 424   |       |
|                                                                                  | RadStd      | 68     | 85    | 141   | 124   | 127   | 143   | 104   | 112   | 111   | 108   | 48    | 39    |       |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (52 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page